

● HARD

● INFORMATION AND IDEAS

● ANSWER KEY

Command of Evidence

10

10 answers · Rationale-rich · Teach from doc

Q1**D**

- D. Mantle-derived rocks younger than 3.2 billion years contain some material that is not found in older mantle-derived rocks but is found in older and contemporaneous lithospheric rocks.

Choice D is the best answer because it presents a finding that, if true, would most directly support the researchers' conclusion that the transition from a stagnant lid regime to a tectonic plate regime occurred around 3.2 billion years ago. The text explains that early in Earth's history, Earth exhibited a stagnant lid regime in which there's no interaction between the lithosphere and the underlying mantle. The text further explains that, by contrast, once Earth began to exhibit a tectonic plate regime, its lithospheric and mantle material began to mix. If mantle-derived rocks younger than 3.2 billion years contain material not found in older mantle-derived rocks, that material must have originated somewhere other than the mantle. And if this material is found in both older and contemporaneous lithospheric rocks, that would imply that the lithosphere was able to mix with mantle material beginning around 3.2 billion years ago, as the researchers concluded.

Choice A is incorrect. The text gives no basis for comparing the quantities of lithospheric and mantle-derived rocks. Choice B is incorrect. The text gives no basis for comparing the material makeup of lithospheric rocks to that of mantle-derived rocks. Choice C is incorrect. A positive correlation between the age of lithospheric rocks and these rocks' chemical similarity to mantle-derived rocks would mean that the oldest rocks would be the most similar, which contradicts the text's claim that lithospheric and mantle-derived rocks were completely separate until 3.2 billion years ago. If the researchers' conclusion about the onset of tectonics on Earth is correct, then younger lithospheric rocks would show greater chemical similarity to mantle-derived rocks than older lithospheric rocks do.

Q2

B

- B. From 2000 to 2006, processing imports with inputs rose much more sharply than processing imports with assembly did.
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Choice B is the best answer because it describes data from the graph that best support the student's assertion that initial efforts at trade liberalization in China were shaped by firms having limited capital (assets available for use) and that this situation resolved during the 2000s. The text explains that an approach to trade liberalization involves engaging in processing imports, one type of which doesn't require payment to a trade partner (processing with assembly) and one type of which requires upfront payment to a trade partner for raw materials (processing with inputs). The graph, which presents China's imports for ordinary imports and both types of processing imports in the years 2000, 2003, and 2006, shows that while processing imports with assembly rose from about 250 hundred million dollars in 2000 to about 750 hundred million dollars in 2006, processing imports with inputs rose much more sharply, increasing from approximately 650 hundred million dollars in 2000 to about 2,300 hundred million dollars in 2006. Because processing with inputs requires firms to pay for materials (expending capital) and processing with assembly doesn't, the sharper rise in processing imports with inputs suggests that Chinese firms' assets—and thus their ability to engage in that type of processing imports—were relatively limited in (and before) 2000 and then substantially increased from 2000 to 2006. In other words, the data suggest that the situation of having limited capital resolved during the 2000s.

Choice A is incorrect because the graph indicates that ordinary imports were greater than both types of processing imports in 2006, not that processing imports with inputs were greater than ordinary imports and processing imports with assembly that year. Choice C is incorrect because the observation that ordinary imports were greater than both types of processing imports in 2000, 2003, and 2006 doesn't address a change within any type of imports from 2000 to 2006, and an indication of a change in that period that might be related to the availability of assets is needed to support the assertion that the situation of having limited capital resolved during the 2000s. Choice D is incorrect because the fact that processing imports with assembly were greater at the end of the period from 2000 to 2006 than processing imports with inputs were at the start of the same period doesn't address a change within either type of imports during the period, and an indication of such a change that might be related to the availability of assets is needed to support the assertion that the situation of having limited capital resolved during the 2000s.

Q3

B

- B. “[Dr. Proudie] was comparatively young, and had, as he fondly flattered himself, been selected as possessing such gifts, natural and acquired, as must be sure to recommend him to a yet higher notice.”
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Choice B is the best answer. In this quotation, Dr. Proudie is described as "fondly flatter[ing] himself" that he has gifts that "must be sure to recommend him to a yet higher notice." In other words, he expects his skills to push him to greater fame and success. This implies an exaggerated sense of his own abilities, which matches the claim we're trying to support.

Choice A is incorrect. This quotation doesn't describe Proudie's view of himself, nor does it paint him in an especially flattering light. Instead, by saying his mental powers and business skill are not "proved," it implies that he is actually dim-witted and bad at business. Choice C is incorrect. This choice describes Proudie's closeness to power and importance, but it doesn't show what Proudie thinks of himself. Proudie is not describing himself or his abilities here. The narrator is. Choice D is incorrect. While this quotation offers Proudie very mild praise, it doesn't show what Proudie thinks of himself or his own abilities, which is what the claim focuses on.

Q4

A

- A. The rules that allow for networks to exhibit behaviors like those of grid cells have no equivalent in the function of biological brains.
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Choice A is the best answer. While many networks can perform navigation tasks, or even mimic grid cells, it doesn't mean they're actually behaving like biological brains—this finding suggests that the rules that govern neural network behavior are completely unlike the way real brains work.

Choice B is incorrect. Although it mentions the rules that are programmed into the networks, this finding wouldn't clarify whether or not these rules have anything to do with the function of biological brains. Choice C is incorrect. This choice suggests that neural networks are modeled after multiple types of brain cells, which sidesteps the question of whether these rule-based networks are genuinely similar to biological brains. Choice D is incorrect. This choice doesn't address the key point of the claim, which is that the apparent similarity between neural networks and biological brains is only due to the rules programmed into the networks. It focuses on training tasks, not the originally programmed rules.

D. The Ultra-Fast Robot Hand weighs only slightly more than the Yale Model T does.

Choice D is the best answer because it describes data in the graph that support Meng and colleagues' conclusion that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T. According to the text, payload capacity is calculated by using a ratio of a robot's holding force to the robot's weight, and higher ratios indicate a greater payload capacity. The Ultra-Fast Robot Hand has a holding force of 56 newtons, four times greater than that of the Yale Model T. Additionally, the graph shows that the Ultra-Fast Robot Hand has a weight of approximately 500 grams, slightly more than the Yale Model T's weight of approximately 480 grams. Therefore, the Ultra-Fast Robot Hand has a higher ratio of holding force to weight than the Yale Model T. Since higher ratios correspond to greater payload capacity, the information from the graph indicating that the Ultra-Fast Robot Hand weighs only slightly more than the Yale Model T combined with the information in the text ultimately supports the conclusion that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T.

Choice A is incorrect. Although, according to the graph, it's true that both the Ultra-Fast Robot Hand and the Yale Model T weigh more than 450 grams, this statement doesn't support Meng and colleagues' conclusion that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T. This statement emphasizes a similarity, not a distinction, between the two robots. Choice B is incorrect. Although, according to the graph, it's true that the Ultra-Fast Robot Hand and the Yale Model T both weigh more than the Permanent Magnet Hand does, this statement doesn't support Meng and colleagues' conclusion that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T. This statement emphasizes a similarity, not a distinction, between the Ultra-Fast Robot Hand and the Yale Model T. Furthermore, the comparison to the Permanent Magnet Hand is irrelevant to the claim about the relative ratios and payload capacities of the Ultra-Fast Robot Hand and the Yale Model T. Choice C is incorrect. Although the text states that the Yale Model T has a lower holding force than the Permanent Magnet Hand, the graph provides no information about holding force. Moreover, information about the Permanent Magnet Hand is irrelevant to the conclusion by Meng and colleagues, which only concerns the Ultra-Fast Robot Hand and the Yale Model T.

Q6

D

- D. As they complete secondary and higher education, former students of the school experience no loss of fluency or cultural pride.
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Choice D is the best answer because it presents a finding that, if true, would support the researchers' prediction about the language nest model of education. The text states that Morcom and Roy studied the effects of the language nest model of education on students at an Anishinaabe school, and they found that the model—which is used with students during pre-K or elementary school—increased students' fluency in the Anishinaabe language and pride in Anishinaabe culture. The researchers predicted that the students' positive early experiences with the Anishinaabe language would lead them to be more likely to later share the language with younger generations. If former students maintain full fluency and cultural pride after finishing secondary and higher education, it follows that they would be both able and motivated to share what they know with others; this would likely result in a higher probability of transmitting the language to younger generations, as the researchers predict.

Choice A is incorrect because finding that Anishinaabe adults who didn't attend the school feel approximately the same degree of cultural pride as those adults who did attend wouldn't support the researchers' prediction that former students will be more likely to share their knowledge with younger generations. This finding would identify a similarity between the groups rather than a factor that might make former students more likely than other adults to transmit the language to younger people. Choice B is incorrect because finding that new students experience increased performance in language fluency and academics would suggest that the school has a positive effect on students when they attended but wouldn't reveal anything about those students' later actions as adults (such as their likelihood of sharing their knowledge with younger generations). Choice C is incorrect because finding that Anishinaabe adults who attended the school are equally likely to stay in the community as adults who didn't attend the school wouldn't support the researchers' prediction that former students will be more likely to share their knowledge with younger generations. This finding would identify a similarity between the groups rather than a factor that might make former students more likely than other adults to transmit the language to younger people.

Q7

D

- D. For each species, the percentage of juvenile plants growing in patches of vegetation was substantially higher than what would be expected if plants were randomly distributed.

Choice D is the best answer because it provides the most direct support from the table for the claim that the plants growing in close proximity to other plants gained an advantage at an early developmental stage. The table shows the total number of juvenile plants from five species that were found growing on bare ground and in patches of vegetation as well as the percentage of the total number of each species that were growing in patches of vegetation. For each of the five species, more than 50% of the juvenile plants were growing in patches of vegetation. The text notes, however, that a random distribution of plants across the landscape should result in only about 15% of the plants being found in patches of vegetation. In other words, for each of the five species, the percentage of juvenile plants found growing in patches of vegetation was substantially higher than could be explained by chance alone. This finding supports the claim in the text: if plants growing in patches are overrepresented among plants that have survived to the juvenile stage, as the data show they are, then it suggests that it's advantageous for plants at an early stage of development to grow in patches of vegetation.

Choice A is incorrect because the statement that less than 75% of juvenile plants were found growing in patches of vegetation, while true, doesn't clearly support the claim that the plants growing in close proximity to other plants gained an advantage at an early developmental stage. Saying that less than 75% of plants were found in patches doesn't indicate how the percentage growing in patches compares with the percentage that would be expected to grow in patches on the basis of chance alone, which is the information necessary to evaluate whether the claim in the text has support in the table. Put another way, if the percentage of plants found growing in patches was 15% or less, it would be true that less than 75% were found in patches, but the data would in fact weaken the claim in the text, not strengthen it, since the data would show that growing in patches wasn't advantageous. Choice B is incorrect because only 12 plants of this species were found growing in patches, which was the lowest number of any species, not the greatest number. Additionally, even if it were true that this species had the greatest number of plants growing in patches, the finding would be irrelevant to the claim that plants of all five species gained an advantage by growing in close proximity to other plants. Choice C is incorrect because 59.1% of the plants of these species were found growing in patches, which is a far greater percentage, not a lower percentage, than what would be expected if plants were randomly distributed (around 15%). Additionally, if it were true that the percentage of plants growing in patches was lower for these species than what would be

expected from chance alone, that finding would weaken, not strengthen, the claim that growing in patches is advantageous.

Q8

B

- B. “Braschi’s eagerness to push boundaries and blend genres within literature invites us to consider how other art forms might also engage with literature.”
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Choice B is the best answer because it presents the quotation that best supports the student’s claim about Braschi. By describing how Braschi’s blending of genres invites her audience to think about how other art forms could also engage with literature, the quotation supports the idea that the diversity of responses to Braschi’s work reflects Braschi’s own approach to creating literature.

Choice A is incorrect because the quotation describes scholars from different countries writing essays about Braschi’s use of language in her writings; it doesn’t address how Braschi’s creation of cross-genre literature inspires diverse types of responses, which is the claim the student makes. Choice C is incorrect because the quotation focuses on the fact that Braschi studied in several different cities, which doesn’t address the student’s claim that Braschi’s creation of cross-genre literature inspires diverse types of responses. Choice D is incorrect because the quotation lists some of the authors who Braschi has written academic works about, which is irrelevant to the student’s claim that Braschi’s creation of cross-genre literature inspires diverse types of responses.

Q9

A

- A. “When constructing his argument regarding the characteristics of a well-functioning government, Aristotle asserts that ‘Transgression creeps in unperceived and at last ruins the state,’ illustrating this idea with a comparison to frequent small expenditures slowly and almost imperceptibly chipping away at a fortune until it is ultimately depleted.”
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Choice A is the best answer because it presents the quotation that best supports the student’s claim that in *The Politics*, Aristotle gives advice on preserving constitutions—preventing governments from falling and maintaining order—and specifically asserts that in a healthy state, laws must be followed as strictly as possible and infractions should not be overlooked even if they are minor. The philosopher states that when Aristotle builds his argument about the characteristics of a well-functioning government, Aristotle asserts that transgression, or violation of law, will ruin the state if it “creeps in unperceived,” or goes unnoticed. The philosopher then adds that Aristotle illustrates this point by comparing the situation to one in which small but frequent expenses diminish a fortune almost unnoticeably until, eventually, the fortune is entirely gone. In other words, the philosopher indicates that Aristotle makes the point that total obedience to law preserves a healthy state while even small violations, if ignored, will undermine the health of the state.

Choice B is incorrect because the philosopher addresses Aristotle’s observation about corruption within the government (in particular, preventing the possibility that members of the government can take bribes), and although corruption can involve infractions, the observation is about a subset of people within the state and isn’t directly connected to the importance of upholding total obedience to the law throughout the state. Choice C is incorrect because the philosopher discusses Aristotle’s point about those who would intentionally destroy a constitution altogether and the need for rulers to remind the populace that it would be dangerous for a constitution to collapse, but neither idea is directly connected to the importance of upholding total obedience to the law. Choice D is incorrect because the philosopher explains that Aristotle makes the point that oligarchic leaders may retain power by having members of disenfranchised classes participate in government alongside governing classes, and this point doesn’t address the importance of ensuring obedience to the law and addressing even minor violations.

Q10

A

- A. Analysis of the animal bones showed that the cattle’s diet also consisted of wheat, which humans widely cultivated in China during the Bronze Age.
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Choice A is the best answer because it presents a finding that, if true, would most strongly support the team’s conclusion that cattle were likely raised closer to human settlements than sheep and goats were. The text explains that Vaiglova, Liu, and their colleagues analyzed the chemical composition of sheep, goat, and cattle bones from the Bronze Age in China in order to investigate the animals’ domestication, or their adaptation from a wild state to a state in which they existed in close connection with humans. According to the text, the team’s analysis showed that sheep and goats of the era fed largely on wild plants, whereas cattle fed on millet—importantly, a crop cultivated by humans. If analysis of the animal bones shows that the cattle’s diet also consisted of wheat, another crop cultivated by humans in China during the Bronze Age, the finding would support the team’s conclusion by offering additional evidence that cattle during this era fed on human-grown crops—and, by extension, that humans raised cattle relatively close to the settlements where they grew these crops, leaving goats and sheep to roam farther away in areas with wild vegetation, uncultivated by humans.

Choice B is incorrect because if it were true that sheep’s and goats’ diets consisted of small portions of millet, which the text states was a crop cultivated by humans, the finding would suggest that sheep and goats were raised relatively close to human settlements, weakening the team’s conclusion that cattle were likely raised closer to those settlements than sheep and goats were. Choice C is incorrect because the finding that cattle generally require more food and nutrients than do sheep and goats wouldn’t support the team’s conclusion that cattle were likely raised closer to human settlements than sheep and goats were. Nothing in the text suggests that cattle were incapable of obtaining sufficient food and nutrients without access to human-grown crops. Hence, even if cattle’s diets are found to have different requirements than the diets of sheep and goats, the cattle could have met those requirements from food located far from human settlements. Choice D is incorrect because if it were true that the diets of sheep, goats, and cattle varied based on what the farmers in each Bronze Age settlement could grow, the finding would weaken the team’s conclusion that cattle were likely raised closer to human settlements than sheep and goats were, suggesting instead that all three types of animals were raised close enough to human settlements to feed on those settlements’ crops.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

